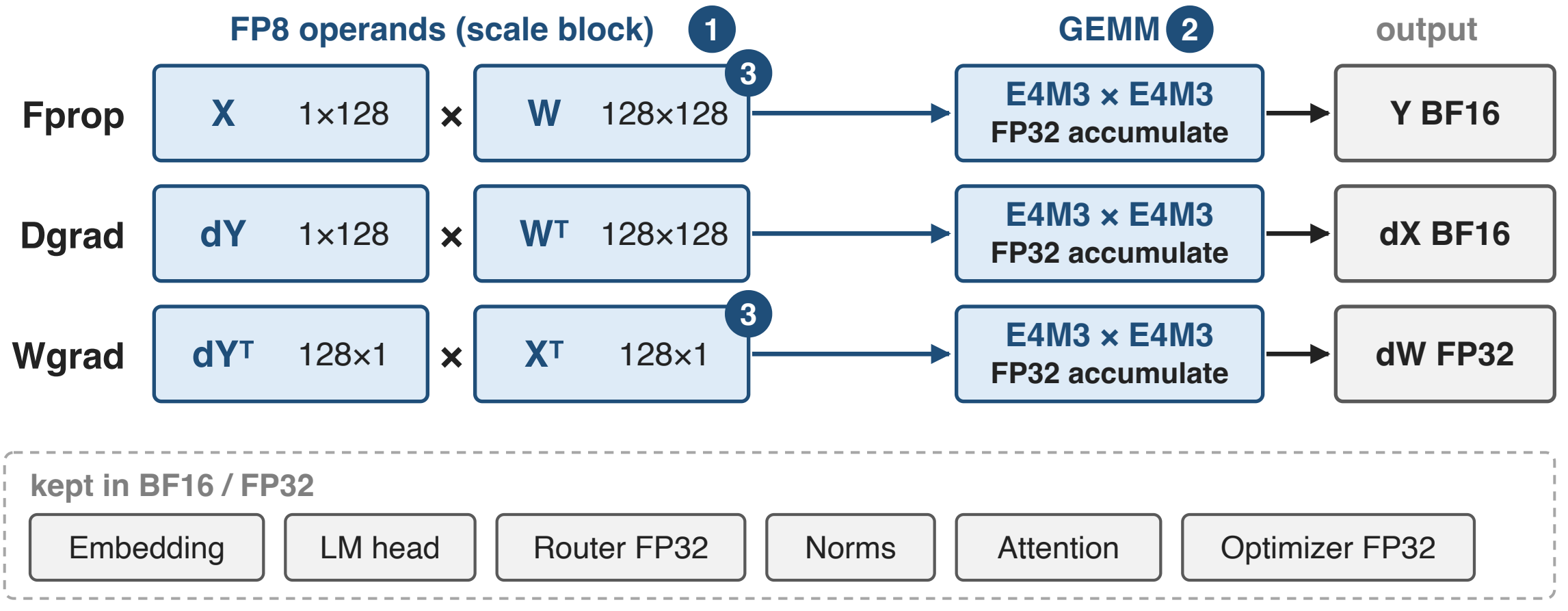

FP8 operands (scale block)
1
GEMM
2
output
Fprop
X 1×128
×
W 128×128
3
E4M3 × E4M3
FP32 accumulate
Y BF16
Dgrad
dY 1×128
×
W^T 128×128
E4M3 × E4M3
FP32 accumulate
dX BF16
Wgrad
dY^T 128×1
×
X^T 128×1
3
E4M3 × E4M3
FP32 accumulate
dW FP32
kept in BF16 / FP32
Embedding
LM head
Router FP32
Norms
Attention
Optimizer FP32